

openmedallion

A simple framework that takes raw data from any source, cleans and transforms it, and turns it into insights — automatically.



THINK OF IT LIKE A KITCHEN



Sources

Grocery store



Bronze

Raw ingredients delivered



Silver

Prepped & measured



Gold

Finished dish, plated



Output

Served to the table

HOW YOU RUN IT

STEP 1

Describe your data in plain files

You write simple

`bronze.yaml`,
`silver.yaml`, `gold.yaml`

config files that say where your data is and what you want done to it. No code required for most tasks.

STEP 2

Run one command

`medallion run <project>`

— that's it. The framework handles the order of operations and logs every step so you know exactly what happened.

STEP 3

Open the results

Every run produces ready-to-use Parquet files your BI team can open directly — Power BI, Tableau, Python notebooks, or anything that reads columnar data.

PYTHON LIBRARIES — WHAT EACH ONE DOES

BRONZE LAYER

`dlt` — Data Load Tool

Connects to databases, files, and APIs. Handles authentication, pagination, and incremental loads — only fetching new rows since the last run. Think of it as a very smart copy-paste that never misses an update.

SILVER & GOLD LAYERS

`Polars` — Lightning-fast tables

Like pandas or Excel formulas but 10–100× faster, written in Rust. Handles joining tables, filtering rows, and computing new columns — all the number-crunching work at each layer.

ALL LAYERS

`Hamilton` — Step orchestrator

Figures out the correct order to run your transformation steps and only re-runs what actually changed. Ensures bronze always runs before silver, silver before gold — automatically.

CONFIG

`YAML` — Human-readable config

All pipeline definitions live in plain YAML files — readable without a programming background. Change a filter, add a column, point to a new database — all without touching Python.

OUTPUT

`Parquet` — Columnar data files

The industry standard for analytical data. Compressed, fast to query, and readable by every modern BI tool — Power BI, Tableau, DuckDB, Python, Spark. No database server needed.

SCHEDULING

`Kestra` — Pipeline scheduler

For recurring scheduled runs (daily, weekly, etc.), openmedallion generates a Kestra flow file automatically. Kestra handles retries, per-task logs, and run history — no cron job hacking.